A finite three-ample complex with a nontrivial fundamental group

Kaiwen Jin
yui.ui.siki@gmail.com

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Abstract

We construct a finite three-ample simplicial complex on 32,768 vertices which retracts simplicially onto a four-cycle. Thus the prescribed extension of every subcomplex on at most three vertices is compatible with a nontrivial fundamental group in a finite complex. The construction fixes a four-cycle and chooses the remaining permitted faces probabilistically. A finite union bound proves that all extension requests can be met at once.

1 The finite existence question

For a simplicial complex X and a set U of its vertices, let $X[U]$ be the induced subcomplex. A nonempty complex is *r-ample* if, for every $U \subseteq V(X)$ with $|U| \leq r$ and every simplicial subcomplex $A \subseteq X[U]$, there is a vertex $v \in V(X) \setminus U$ such that

$$\mathrm{Lk}_X(v) \cap X[U] = A. \tag{1}$$

The subcomplex A may be empty and need not be induced. We regard complexes as families of nonempty simplices, so an empty subcomplex has no simplices; one may equivalently adjoin the formal empty face consistently.

Ampleness requires every prescribed local pattern, including missing higher faces, to admit an extension. Four-ampleness implies simple connectivity [2, Proposition 6.1], whereas Barmak constructed an infinite three-ample complex which is not simply connected [1, Theorem 1]. Farber asks whether a finite example exists [2, Example 2.5]. We answer this finite existence question.

Theorem 1. *There is a finite three-ample simplicial complex X with* 32,768 *vertices and dimension at most three which admits a simplicial retraction onto a four-cycle. In particular, $\pi_1(|X|)$ contains a subgroup isomorphic to $\mathbb{Z}$.*

The four-cycle occurs in [1, Example 6]. We replace each of its vertices by a large set of vertices and form an ambient complex whose simplices project onto vertices or edges of the cycle. We then choose a random subcomplex while retaining a fixed copy of the cycle. The projection onto the cycle is therefore available in every outcome. Any three labels omit a fourth label, leaving a whole set of vertices which can realize the required local extension. The stated number of vertices is a convenient choice for the probability estimate; no minimum-size claim is made.

2 Construction and proof

Proof of Theorem 1. Let C be the four-cycle with vertex set $\mathbb{Z}/4\mathbb{Z}$ and edges $\{i, i+1\}$, where indices are read modulo four. Put $m = 8192$ and

$$V = (\mathbb{Z}/4\mathbb{Z}) \times \{0, \dots, m-1\}, \qquad N = 4m.$$

Write $p(i,j) = i$ and $s(i) = (i, 0)$. A nonempty set $\tau \subseteq V$ is *permitted* if $|\tau| \le 4$ and $p(\tau)$ is a simplex of C; thus its distinct labels form a singleton or an adjacent pair. These sets form a simplicial complex B of dimension at most three. The four edges $\{s(i), s(i+1)\}$ form a subcomplex S isomorphic to C.

For each permitted τ with $2 \le |\tau| \le 4$, choose a mark $b_\tau \in \{0, 1\}$. Set the four marks on the edges of S equal to one and choose every other mark independently, with probabilities $1/2$ for each value. Include all vertices in X, and include a permitted simplex τ of size at least two precisely when

$$b_\rho = 1 \quad \text{for every } \rho \subseteq \tau \text{ with } |\rho| \ge 2. \tag{2}$$

This rule is closed under taking faces, so X is a finite subcomplex of B with all N vertices. It contains S in every outcome. Both $p : X \to C$ and $s : C \to X$ are simplicial, and $p \circ s = \mathrm{id}_C$.

We now show that some outcome is three-ample. Fix $U \subseteq V$ with $1 \le |U| \le 3$. Choose a label $q \notin p(U)$, and put $t = q + 2$. Then $p(U) \subseteq \{t-1, t, t+1\}$. Every face σ of $X[U]$ has one label or an adjacent pair of labels in this set; the only such pairs are $\{t-1, t\}$ and $\{t, t+1\}$. It follows that $p(\sigma) \cup \{t\}$ is a simplex of C. Consequently $\{v\} \cup \sigma$ is permitted whenever $p(v) = t$ and $v \notin U$.

Let W consist of the vertices with label t outside U and outside the four designated vertices $s(i)$. There are at least $m - 4$ such vertices. Condition on $X[U] = K$, where K has positive probability, and fix any subcomplex $A \subseteq K$. For $v \in W$, prescribe

$$b_{\{v\}\cup\sigma} = \begin{cases} 1, & \sigma \in A, \\ 0, & \sigma \in K \setminus A, \end{cases} \qquad \sigma \in K. \tag{3}$$

Call this event E_v. There are $f(K)$ prescribed marks, where $f(K)$ is the number of nonempty simplices of K, and $f(K) \le 7$.

The event $X[U] = K$ depends only on the marks of faces contained in U. Every mark in (3) contains the vertex $v \notin U$ and is unfixed, since v is not a designated vertex. For different vertices $v, w \in W$, these mark sets are disjoint: a set indexing a prescribed mark has exactly one vertex outside U, namely its own candidate vertex. Thus, after conditioning on $X[U] = K$, all prescribed marks remain fair and the events E_v are mutually independent. This can also be seen by counting completions of any assignment of the marks inside U: prescribing d additional coordinates leaves a proportion 2^{-d} of the completions. In particular,

$$\mathbb{P}(E_v \mid X[U] = K) = 2^{-f(K)} \ge \frac{1}{128}. \tag{4}$$

On E_v, the restricted link is exactly A. Indeed, if $\sigma \in A$, all its nonempty faces belong to A. All marked faces of $\{v\} \cup \sigma$ not containing v already have mark one, because $\sigma \in K$. Those containing v have mark one by (3), so (2) includes $\{v\} \cup \sigma$ in X. If $\sigma \in K \setminus A$, its own cone mark is zero, and this cone face is excluded. This proves (1), including empty A and subcomplexes with omitted edges or triangles.

Set $r = 127/128$. Conditional independence and (4) give

$$\mathbb{P}(A \text{ has no outside witness} \mid X[U] = K) \le r^{|W|} \le r^{m-4}.$$

There are at most $2^{f(K)} \le 128$ subcomplexes of the fixed K. Taking their union first, and then averaging over the possible induced complexes K, shows that the probability of any failed request for this U is at most $128 r^{m-4}$. For $U = \varnothing$, the sole request is automatic since V is nonempty. A union bound over every U of size at most three therefore yields

$$\mathbb{P}(X \text{ is not three-ample}) \le 128 \sum_{k=0}^{3} \binom{N}{k} r^{m-4} \le 512 N^3 r^{m-4}. \tag{5}$$

Here $\binom{N}{k} \le N^k \le N^3$ for $0 \le k \le 3$. The binomial theorem gives

$$r^{-128} = \left(1 + \frac{1}{127}\right)^{128} \ge 1 + \frac{128}{127} > 2, \qquad r^{128} < \frac{1}{2}.$$

Since $m - 4 = 8188 > 128 \cdot 63$ and $N = 2^{15}$, the last bound in (5) is strictly less than

$$2^{54} (r^{128})^{63} < 2^{54} 2^{-63} = 2^{-9} < 1.$$

The probability space is finite. Hence it has an outcome X satisfying every extension request at once.

This X is path connected: any two vertices have a common neighbor by using the subcomplex consisting of those two vertices in (1), and each point of a simplex can be joined to one of its vertices inside that simplex. The fixed maps s and p still satisfy $p \circ s = \mathrm{id}_C$. Since $|C|$ is a circle, functoriality of fundamental groups, with basepoints 0 and $s(0)$, gives

$$p_* \circ s_* = \mathrm{id}_{\pi_1(|C|)}, \qquad \pi_1(|C|) \cong \mathbb{Z}$$

[3, Theorem 1.7 and p. 34]. Thus s_* is injective, proving the fundamental-group assertion. $\square$

AI use

OpenAI Codex assisted in developing the argument, checking references and preparing this manuscript. Independent mathematical review is requested.

References

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